DEVOPS AND CLOUD-BASED SOFTWARE PROJECT:

Migrate Service To The Cloud

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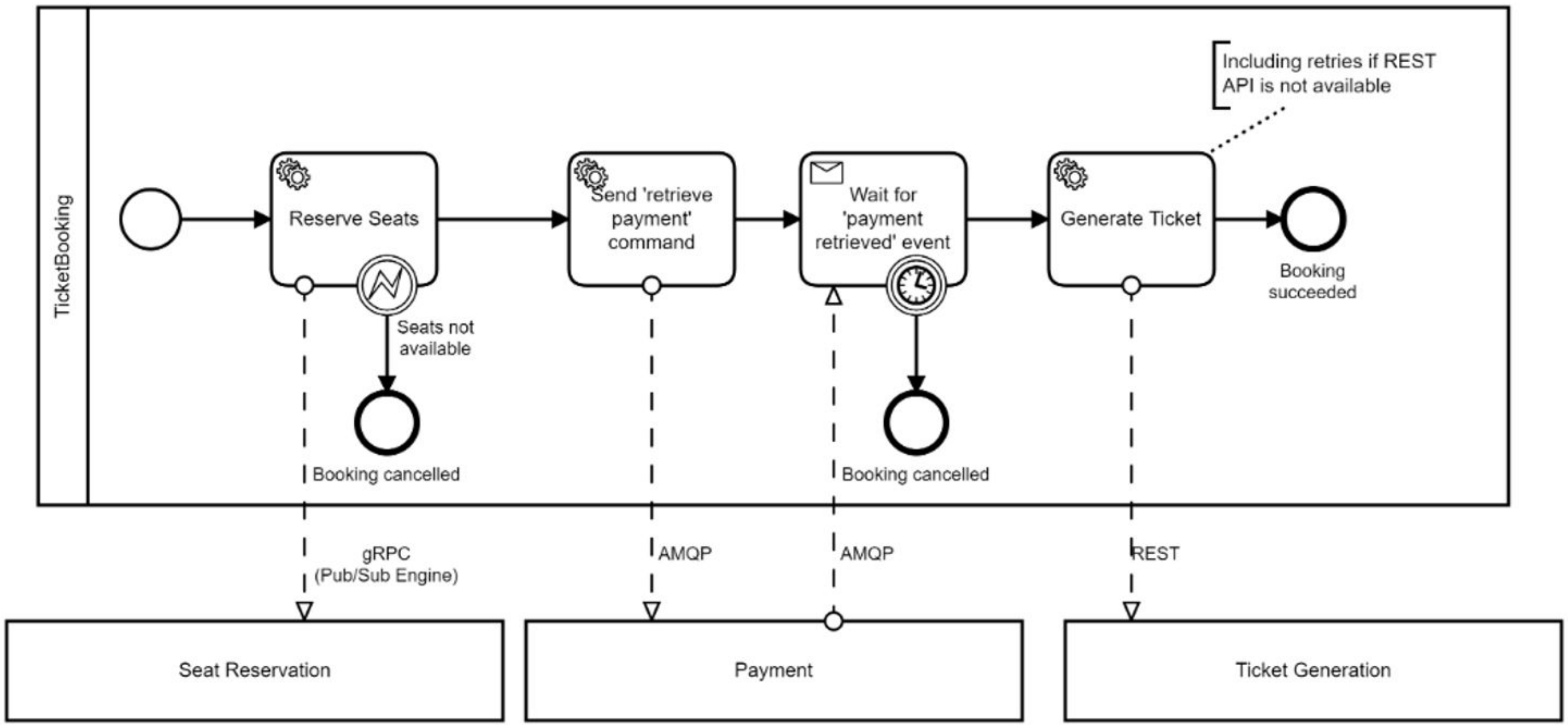
Ticket Booking App



https://github.com/berndruecker/ticket-booking-camunda-8

Project Goal: From Local Workflow to Cloud-Native System

- ➔ Move the existing system to the **cloud**
- ➔ Keep the **core business logic unchanged**
- ➔ Make the system:
  - ◆ Scalable
  - ◆ Observable
  - ◆ reliable
- ➔ Apply **DevOps practices**:
  - ◆ CI/CD
  - ◆ Automation
  - ◆ monitoring
- ➔ Evaluate **cost and performance trade-offs**

Week 2 – Project Kickoff & Architecture (Planning Sprint)

Understand the project and make architectural decisions.

Expected Deliverables

- High-level **architecture diagram** showing (cloud provider/compute model/tool/service):
  - workflow engine
  - services
  - communication mechanisms
  - database / storage
  - monitoring & alerting
- GitHub repository created
  - basic README describing the architecture

Week 3 – First Cloud Deployment (CD foundation)

Get something running in the cloud.

Expected Deliverables

- Application deployed in the cloud (basic)
- CI pipeline exists:
  - push → build → deploy
- Public endpoint works (e.g. `/ticket`)
- Basic logs visible

Week 4 – Autoscaling & Load Handling (Elasticity Sprint)

Demonstrate elasticity under load.

Expected Deliverables

- Autoscaling enabled
- Load test executed
- Scaling behavior observed

Week 5 – Monitoring, Alerts & Anomaly Detection (Ops Sprint)

Operate the system like a production service.

Expected Deliverables

- Monitoring dashboard:
  - latency
  - error rate
  - resource usage
- (Optional advanced) At least 2 alerts, e.g.:
  - CPU > threshold
  - latency spike
  - error rate anomaly
- Explanation of normal vs abnormal behavior

Week 6 – Cost, Optimization & Final Demo (Hardening Sprint)

Reflect, optimize, and finalize.

Expected Deliverables

- Cost estimation:
  - compute
  - scaling impact
- (Optional advanced)
  - performance optimization
  - anomaly detection beyond thresholds
- Final demo + short reflection